

ZigBee network model does not use presentation, session or transport layer, and the user application is directly tied to the application layer (APL). The APL consists of the application sub-layer (APS), ZigBee device object (ZDO) and application framework.

The ZigBee IEEE802.15.4 standard defines three frequency bands to employ a standard all over the world. Out of these three frequencies, the 2.45GHz platform is commonly used at 250kbps data transfer rate. Secure data transfer is possible with the help of secure service provider. The common medium-access control (MAC) layer provides a variety of wireless LAN functions. It manages and maintains communication between the radio network and access points by coordinating.

ZigBee standard devices

The ZigBee devices that incorporate ZigBee radio are:

1. Coordinator: Organises the network and maintains routing tables
2. Routers: These can talk to the coordinator, other routers and reduced-function end devices
3. End devices: These can talk to routers and the coordinator, but not to each other

The coordinator and routers are defined as full-function devices, which are powered-on all the time when mains power is connected. The end devices are reduced-function devices having a shorter protocol stack without the ability of routing but these devices could be battery-powered.

Sensors and actuators could be connected to each of these three ZigBee devices. Except commonly used mesh topology, it is also possible to use tree or star topology, which requires less hardware and software resources of the microcontroller.

The ZigBee device object defines the role of the device (coordinator, router or end device) in the network, creates or answers the connection requests and forms security connection between ZigBee devices. It

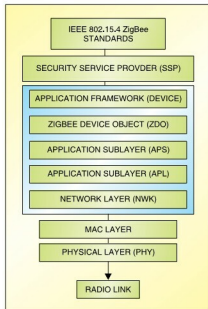


Fig. 1: The ISO-OSI network reference model for ZigBee purposes

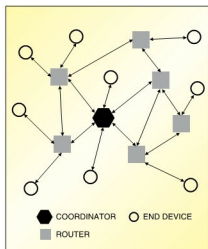


Fig. 2: ZigBee devices include coordinator, end devices and routers

discovers new devices in the network and investigates their offered services as well.

The main task of the network layer is securing of frames and routing to end nodes. It finds direct neighbour nodes and saves information about them. The coordinator network layer ensures communication and distributes addresses to newly connected devices.

The MAC layer synchronises the network, ensures access to the radio channel, checks validity of the frame,

acknowledges reception of the frame, controls connection, and generates and recognises addresses.

The application sub-layer takes care of binding tables, which allow matching (binding) of devices according to their offered services. Communication between devices depends on ZigBee device profile—summary of device properties—and definition of message formatting so that the devices create practical application.

ZigBee-based home automation

With the help of ZigBee module, one can implement a wireless sensor network for home automation. Fig. 3 shows the block diagram of the proposed home automation network. It consists of a PIC controller as the central processing unit. The signals sensed by the ZigBee receiver unit are given, through the PIC controller, to the relay circuit. The relay circuit, in turn, allows various controlling actions to take place.

The working of sensors follows:

Light sensor network. When daylight falls on the light sensor, the ZigBee transmitter unit transmits daylight signals to the ZigBee receiver unit. The received signals are fed to the controller unit, which, in turn, switches off all the lights of the house via the relay circuit.

At dusk, when daylight falling on the sensor drops to a low level, the transmitter unit transmits another signal to the receiver. The sensed signal is fed to the controller, which, in turn, drives the relay circuit to switch on the lights of the house.

Smoke sensor network. When smoke or fire is sensed by the sensor, the controller unit activates the alarm circuit.

Temperature sensor network. The temperature sensor senses the temperature of the room. If the temperature goes above the preset value, the controller unit drives the relays to switch on the fans/AC. When the temperature falls below the preset value, the fans/